QuRNG

A Hardware Agnostic Quantum Random Number Generator

Highlights:

Transparent design
Secured
Flexible and easy-to-use
Confidential
Test suites qualified
Quantum hardware agnostic
Accessible with any smart device
Adaptable to applications
Scalable architecture

Random numbers form the bedrock of security and privacy in our digitally dependent era. The quest for true randomness has driven numerous advancements in Random Number Generator (RNG) technology. Quantum Random Number Generators (QRNGs) harness the inherent randomness of quantum mechanics. For instance, a single photon passing through a beam splitter exhibits unpredictable behavior according to quantum laws, resulting in truly random outcomes. QRNGs offer unparalleled promise for ensuring quantum-secure communications and data protection. However, current QRNGs are hardware-based, requiring dedicated quantum hardware, and users must trust vendors for the quality of random bits generated.

Applications:



Network
Routing Protocols



Online Gaming
and Lotteries



Identity and Access
Management



Modeling of
Financial Systems



Risk Management
and Option Pricing



Secure Data Storage
and Digital Assets



Secure Token and OTP
Generation for Digital Payments



Heuristic Algorithm based
Simulations/ Monte Carlo Simulations

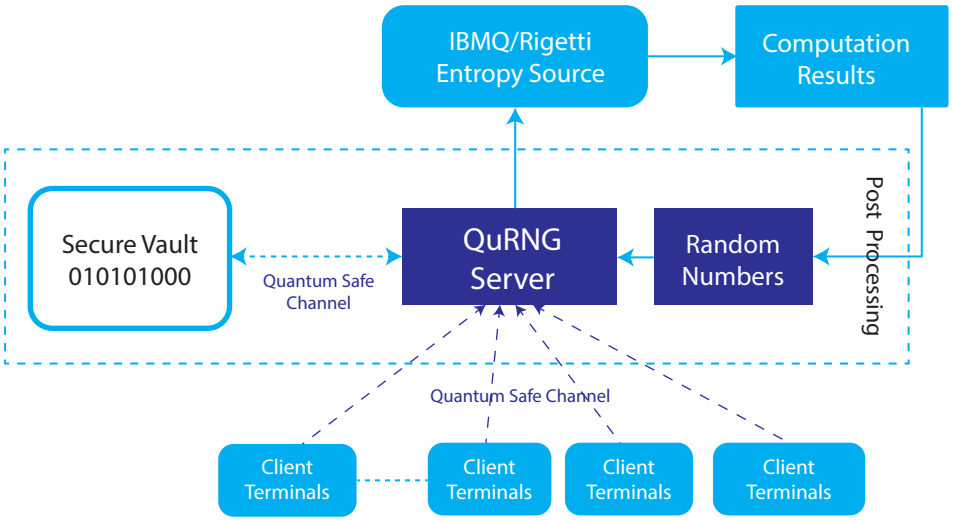


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QuRNG

QuRNG is a revolutionary product developed by QAIG that overcomes accessibility barriers and guarantees high-quality random bits anytime and anywhere from various quantum hardware sources. Its unique design eliminates reliance on dedicated quantum hardware, simplifying software integration. Users can select their entropy source and run multiple QRNGs in parallel. QuRNG™ is a hybrid Quantum-Classical software utilizing a quantum subroutine compatible with any gate-based quantum computer to generate random numbers. It combines the high-quality randomness of hardware QRNGs with the scalability and ease of integration found in software-based Pseudorandom Number Generators (PRNGs), ensuring robust, scalable, and easily integrable random number generation.

System Architecture:



Specifications:

Features	QuRNG
Bit rate	>15 Mbit/s
NIST and Diehard Test Compliance	Yes
Quantum Tests	CHSH Bell Test Qualified
Scalability	Easily scalable with software updates
External Dependencies	None
Requirement	Internet connectivity
Compatible Quantum Computers	IBMQ , Rigetti
Channel Security	Post-Quantum Cryptography Enabled
Security Vault	Hosted in house with Post-Quantum Encryption
Post Quantum Encryption	Quantum wrapper channel for secure key transmission
SaaS	Yes
Supported OS	Linux , Windows

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